# gov51-ps2

Angela Calderon Pio

https://github.com/angelacalderonpio/gov51-ps2.git

## Gov 51 Problem Set 2

## 1.1 Exploring the Experiment

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(tinytex)
```

```
Warning: package 'tinytex' was built under R version 4.5.2
```

```r
gay <- read_csv("data/raw/gay.csv")
```

```
Rows: 69592 Columns: 4
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): treatment
dbl (3): study, wave, ssm

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
##Q1.1

nrow(gay)
```

```
[1] 69592
```

```r
gay |>
  group_by(study) |>
  summarize(n = n())
```

```
# A tibble: 2 x 2
  study     n
  <dbl> <int>
1     1 59207
2     2 10385
```

```r
gay|>
  group_by(treatment) |>
  summarize(n = n())
```

```
# A tibble: 5 x 2
  treatment                                    n
  <chr>                                    <int>
1 No Contact                               37735
2 Recycling Script by Gay Canvasser         6514
3 Recycling Script by Straight Canvasser    6474
4 Same-Sex Marriage Script by Gay Canvasser      12452
5 Same-Sex Marriage Script by Straight Canvasser  6417
```

There are 69,592 total observations in gay.csv. The study has 2 unique values (1 or 2). There are 5 treatment conditions, which are: "No Contact," "Recycling Script by Gay Canvasser," "Recycling Script by Straight Canvasser," "Same-Sex Marriage Script by Gay Canvasser," or "Same-Sex Marriage Script by Straight Canvasser."

```
##Q1.2

gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment                                      mean_ssm
  <chr>                                             <dbl>
1 No Contact                                         3.04
2 Recycling Script by Gay Canvasser                  3.13
3 Recycling Script by Straight Canvasser             3.01
4 Same-Sex Marriage Script by Gay Canvasser          3.03
5 Same-Sex Marriage Script by Straight Canvasser     3.10
```

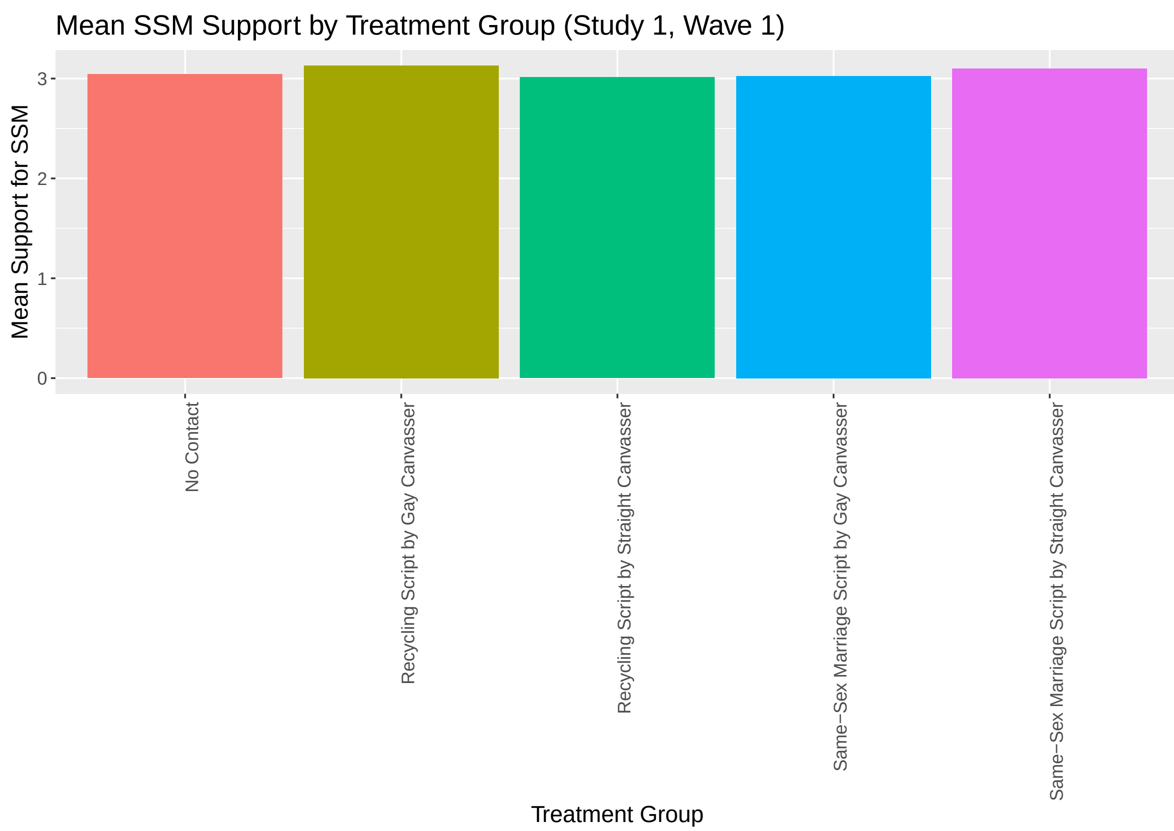
The treatment group that shows the highest average support is Recycling Script by Gay Canvasser.

```
##Q1.3

gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment                                      sd_ssm
  <chr>                                           <dbl>
1 No Contact                                       1.68
2 Recycling Script by Gay Canvasser                1.70
3 Recycling Script by Straight Canvasser           1.68
4 Same-Sex Marriage Script by Gay Canvasser        1.67
5 Same-Sex Marriage Script by Straight Canvasser   1.68
```

The standard deviations are roughly similar across the five treatment groups, ranging from roughly 1.66 to 1.70. This similarity is important because in a randomized experiment it suggests that the groups have comparable variability in responses as they are being drawn from the same underlying population and randomly assigned to treatments. If the variability were very different among the groups, it could indicate that the groups were not drawn from comparable populations which would make it more difficult to attribute differences in outcome to the treatment.

```
##Q1.4

bar_plot_gay<- gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

ggplot(bar_plot_gay, aes(x= treatment, y= mean_ssm,
                         fill = treatment)) +
  geom_col() +
  labs(x = "Treatment Group", y = "Mean Support for SSM",
       title = "Mean SSM Support by Treatment Group (Study 1, Wave 1)") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  theme(legend.position = "none") +
  theme(text = element_text(size =14))
```




```
##Q1.5
q1_data <- gay |>
  filter(study ==1, wave ==1)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data |>
  group_by(treatment) |>
summarize(mean_ssm = mean(ssm, na.rm = TRUE), sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 1 x 3
  treatment  mean_ssm sd_ssm
  <chr>         <dbl>  <dbl>
1 No Contact     3.04   1.68
```

```
q1_data
```

```
# A tibble: 5,238 x 4
```

```
   study treatment   wave   ssm
   <dbl> <chr>      <dbl> <dbl>
 1     1 No Contact     1     5
 2     1 No Contact     1     4
 3     1 No Contact     1     5
 4     1 No Contact     1     4
 5     1 No Contact     1     5
 6     1 No Contact     1     3
 7     1 No Contact     1     2
 8     1 No Contact     1     5
 9     1 No Contact     1     4
10     1 No Contact     1     1
# i 5,228 more rows
```

```r
(3.025195 - 3.042764) / sqrt((1.666099^2 + 1.680540^2)/2)
```

```
[1] -0.01049939
```

The standardized difference is roughly -0.01, which is much smaller than 0.25 in absolute value, suggesting that the "Same-Sex Marriage Script by Gay Canvasser" group and the "No Contact" group are well balanced at baseline. Therefore, the groups appear similar before treatment.

```r
##Q1.6
q1_data2 <- gay |> filter(study ==1, wave ==2)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data2 |>
  group_by(treatment) |>
summarize(mean_ssm = mean(ssm, na.rm = TRUE))
```

```
# A tibble: 1 x 2
  treatment  mean_ssm
  <chr>         <dbl>
1 No Contact     3.04
```

```r
(3.139489   - 3.039615)
```

```
[1] 0.099874
```

The difference in means is 0.099874, meaning the gay canvasser group has about 0.10 higher support for same-sex marriage than the no contact group in Wave 2. Because the groups were balanced at baseline, this suggests that this difference at Wave 2 likely reflects the effects of the canvassing treatment (average treatment effect). In other words, this suggests that exposure to the gay canvasser increased support for same-sex marriage relative to no contact.

## 2.1 The Baseline Distribution

```r
##Q2.1
gay_r <- read_csv("data/raw/gayreshaped.csv")
```

```
Rows: 11948 Columns: 6
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): treatment
dbl (5): study, therm1, therm2, therm3, therm4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ccap <- read_csv("data/raw/ccap2012.csv")
```

```
New names:
Rows: 43998 Columns: 3
-- Column specification
-------------------------------------------------------- Delimiter: "," dbl
(3): ...1, caseid, gaytherm
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```r
nrow(gay_r)
```

```
[1] 11948
```

```r
ncol(gay_r)
```

```
[1] 6
```

```r
colnames(gay_r)
```

```
[1] "study"     "treatment" "therm1"    "therm2"    "therm3"    "therm4"
```

```r
nrow(ccap)
```

```
[1] 43998
```

```r
ncol(ccap)
```

```
[1] 3
```

```r
colnames(ccap)
```

```
[1] "...1"     "caseid"   "gaytherm"
```

For Gay Reshaped, there are 11,948 observations and 6 columns: "study," "treatment," "therm1," "therm2," "therm3," and "therm4."

For Ccap, there are 43,998 observations and 3 columns: "…1," "caseid," and "gaytherm."

```r
##Q2.2

gay_r |>
  filter(study ==1) |>
  summarize(mean = mean(therm1, na.rm= TRUE),
            median = median(therm1, na.rm = TRUE),
            sd = sd(therm1, na.rm = TRUE))
```

```
# A tibble: 1 x 3
   mean median    sd
  <dbl>  <dbl> <dbl>
1  58.4     52  28.5
```

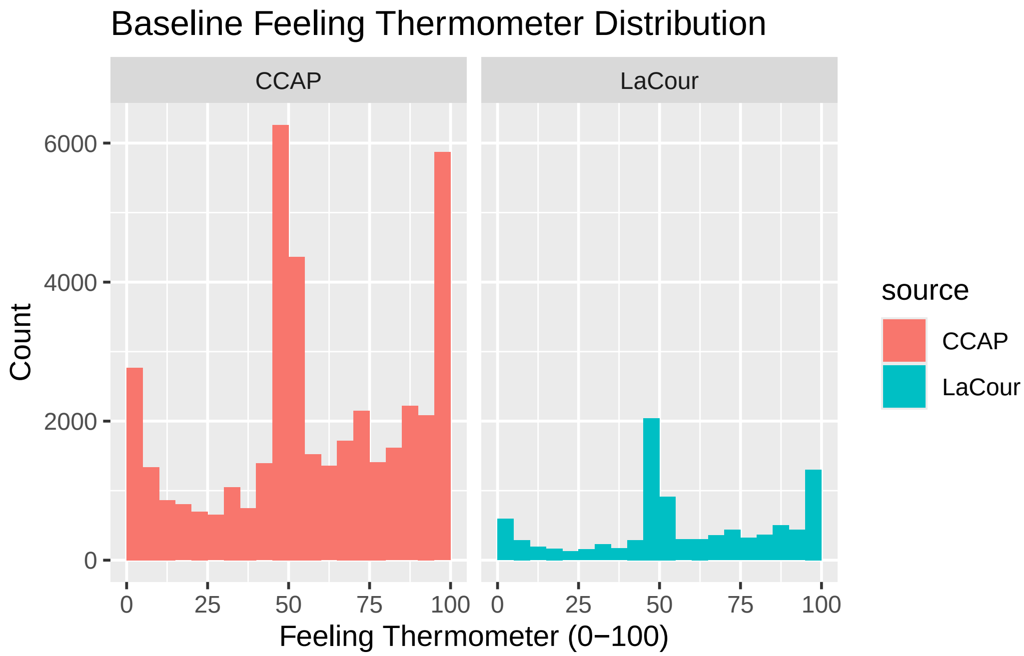
```r
ccap|>
  summarize(mean = mean(gaytherm, na.rm= TRUE),
            median = median(gaytherm, na.rm = TRUE),
            sd = sd(gaytherm, na.rm = TRUE))
```

```
# A tibble: 1 x 3
   mean median    sd
  <dbl>  <dbl> <dbl>
1  58.7     54  29.4
```

I notice that the summary statistics of the two datasets are very similar. The means, medians, and standard deviations are nearly the same.

```
#Q2.3

lacour <- gay_r |>
  filter(study ==1) |>
  select(therm1) |>
  mutate(source = "LaCour")

ccap_data <- ccap |>
  select(gaytherm) |>
  rename(therm1 =gaytherm) |>
  mutate(source = "CCAP")

both <- bind_rows(lacour, ccap_data)

ggplot(both, aes(x=therm1, fill = source)) +
  geom_histogram(breaks = seq(0,100, by=5)) +
                 facet_wrap(~source) +
  labs(title = "Baseline Feeling Thermometer Distribution",
       x = "Feeling Thermometer (0-100)",
       y = "Count")
```

```
Warning: Removed 3097 rows containing non-finite outside the scale range
(`stat_bin()`).
```




The two histograms appear roughly similar in their overall distribution. For example, both show concentrations and the lower and upper ends of the scale and a large share of responses around 50 on the x-axis. Because the shapes look quite similar, it does not appear that the two datasets come from very different populations.

```
#Q2.4

#Study 1 control group

study1_control <- gay_r |>
  filter(study ==1, treatment == "No Contact")

cor(study1_control$therm1, study1_control$therm2, use = "complete.obs")
```

```
[1] 0.9975817
```

```
sd(study1_control$therm2 - study1_control$therm1, na.rm = TRUE)
```

```
[1] 1.987678
```

```r
## Study 2 control group

study2_control <- gay_r |>
  filter(study ==2, treatment == "No Contact")

cor(study2_control$therm1, study2_control$therm2, use = "complete.obs")
```

```
[1] 0.9734449
```

```r
sd(study2_control$therm2 - study2_control$therm1, na.rm = TRUE)
```

```
[1] 6.605738
```

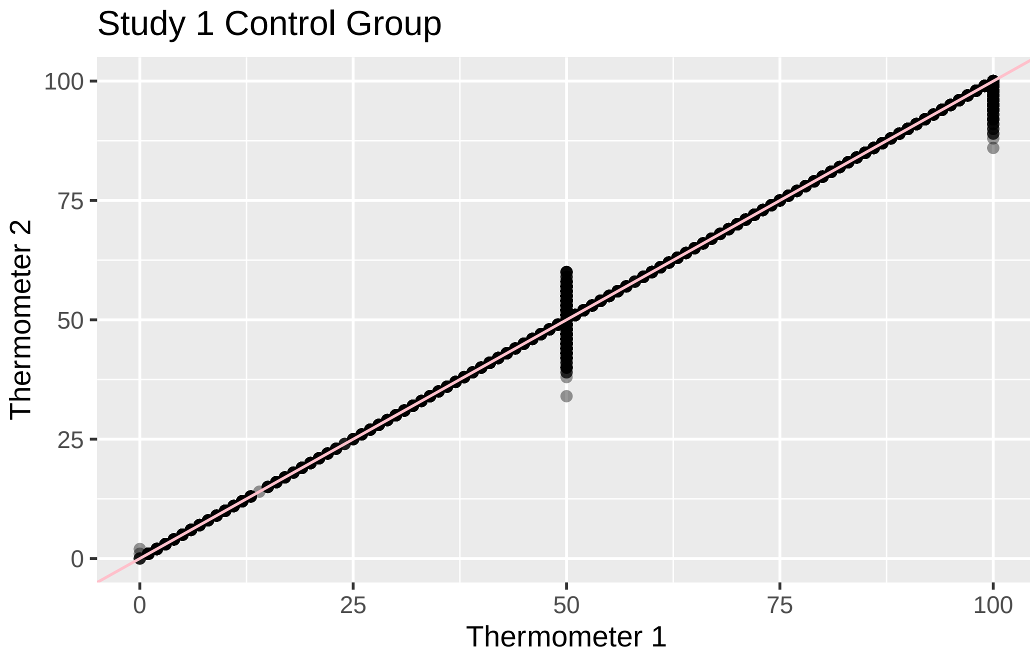
Study 1: correlation = 0.9976, SD of change = 1.99

Study 2: correlation = 0.9734, SD of change = 6.61

Study 2 demonstrates a more plausible pattern because its correlation found in this problem (0.973) is within the typical range of 0.85-0.97 for real feeling thermometer data. On the other hand, Study 1's correlation (0.9976) is very close to 1. Furthermore, taking a look at the standard deviation, Study 2's (6.61) is much larger than Study 1's (1.99): this could point to a more realistic variation in responses. Therefore, Study 2 seems more believable.

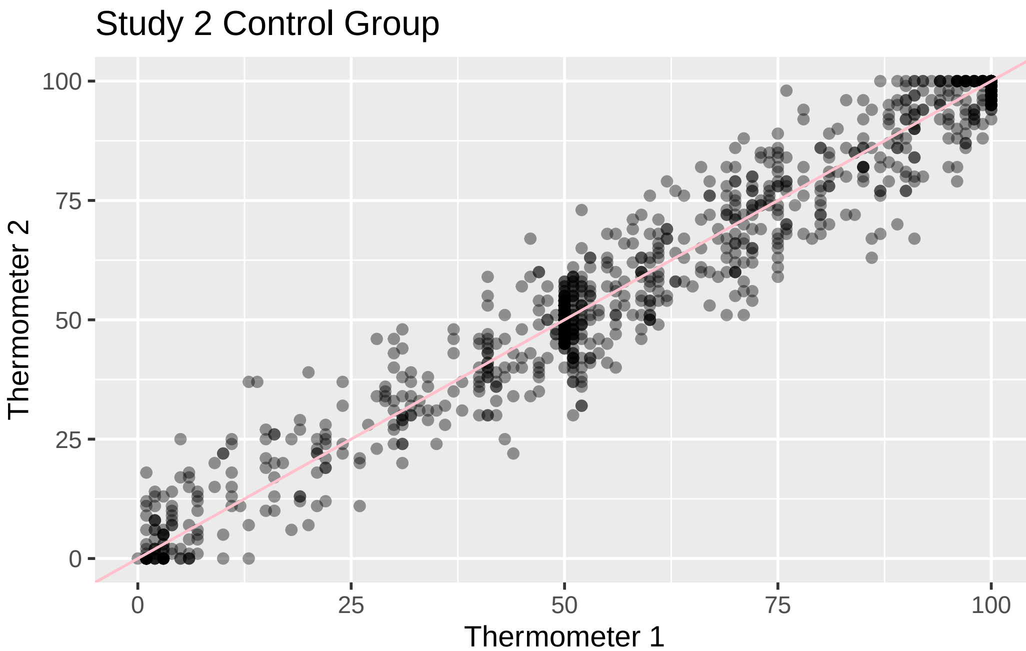
```r
#Q2.5

ggplot(study1_control, aes(x= therm1, y=therm2)) +
  geom_point(alpha = 0.4) +
  geom_abline(slope = 1, intercept = 0, color = "pink") +
  labs(title = "Study 1 Control Group",
       x = "Thermometer 1", y = "Thermometer 2")
```

```
Warning: Removed 568 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
ggplot(study2_control, aes(x= therm1, y=therm2)) +
  geom_point(alpha = 0.4) +
  geom_abline(slope = 1, intercept = 0, color = "pink") +
  labs(title = "Study 2 Control Group",
       x = "Thermometer 1", y = "Thermometer 2")
```

```
Warning: Removed 164 rows containing missing values or values outside the scale range
(`geom_point()`).
```




Study 2 looks more like real longitudinal data. The Study 1 scatter plot depicts the points fitting almost perfectly along the 45 degree line, with the exception of around x= 0, 50, and 100, meaning that the responses collected during Wave 2 are nearly the same to those at Wave 1. On the other hand, Study 2's plot demonstrates more spread around the line, meaning that respondents' answers changed somewhat between the two waves. This pattern is more realistic an consistent with actual longitudinal survey data because it's more realistic that responses vary over time.

### Q2.6

The findings discovered in question 2 suggest that perhaps the data was fabricated rather than collected in a real experiment. LaCour's findings were suspiciously similar to that of CCAP's national survey. For example, the baseline distributions in Study 1 closely match those of the national survey, which again, should signal some alarm because Lacour claimed to have used his own sample. The shape of the two thermometer histograms are quite similar. Secondly, the test correlation in Study 1 is extremely higher (at roughly 0.998), when typically real feeling thermometer data is around 0.95 to 0.97. And then, the standard deviation of the within-person change was about 1.99, which means that respondents' answers hardly changed in between waves, which isn't realistic because people's mood changes often and can impact their responses. A fabricator might produce data like this because consistent responses realistic-looking baseline distributions can make the results appear convincing.

### 3.1 Comparing Baselines

```
gay_r |>
  filter(study ==2) |>
  nrow()
```

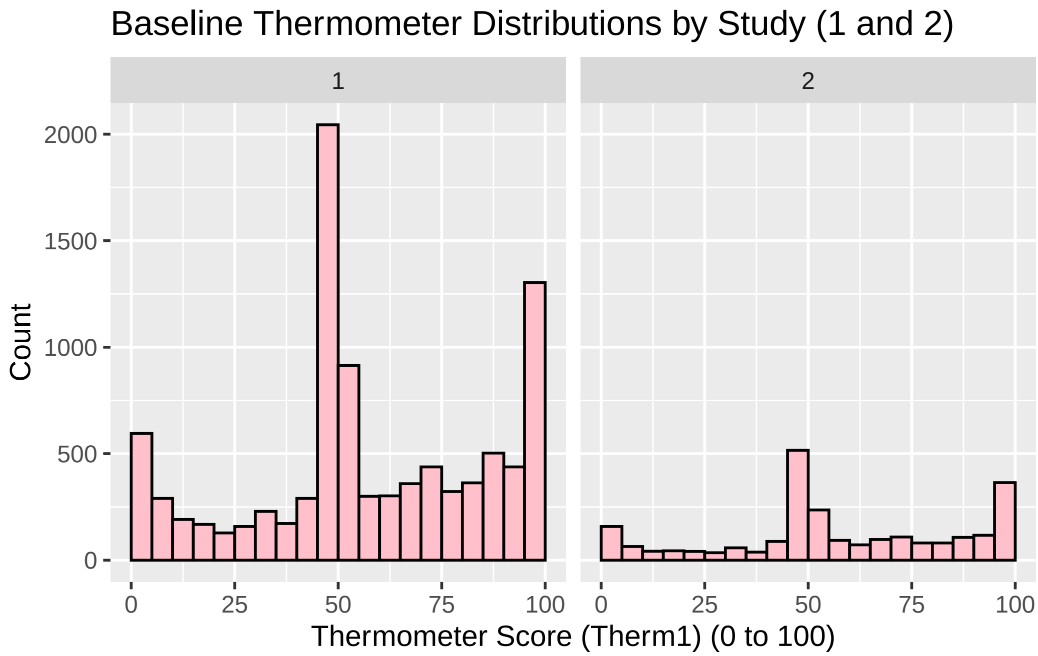
```
[1] 2441
```

```
gay_r |>
  filter(study ==2) |>
  group_by(treatment) |>
  summarize(
    mean_therm1 = mean(therm1, na.rm= TRUE),
            sd_therm1 = sd(therm1, na.rm = TRUE))
```

```
# A tibble: 2 x 3
  treatment                                  mean_therm1 sd_therm1
  <chr>                                            <dbl>     <dbl>
1 No Contact                                        57.9      28.7
2 Same-Sex Marriage Script by Gay Canvasser         59.4      28.4
```

There are 2,441 observations in Study 2. The mean and standard deviation for the No Contact group is 57.89 and 28.70, respectively. For Same-Sex Marriage Script with the Gay Canvasser treatment, the mean was 59.35 and 28.42, respectively.

```
#Q3.2

gay_r |>
  filter(study %in% c(1,2)) |>
  ggplot(aes(x=therm1)) +
  geom_histogram(breaks = seq(0,100,5), fill = "pink", color = "black") +
  facet_wrap (~study) +
  labs(title = "Baseline Thermometer Distributions by Study (1 and 2)",
       x = "Thermometer Score (Therm1) (0 to 100)",
       y = "Count")
```




The Baseline distributions for Study 1 and Study 2 look similar in shape. For example, both graphs display clustering around 0, 50, and 100. Study 1 appears more concentrated around certain values, whereas Study 2 seemingly depicts more of a spread.

```
#Q3.3

q33<- gay_r |>
  filter(study ==2)|>
filter(treatment %in% c(
"No Contact", "Same-Sex Marriage Script by Gay Canvasser"))

q33 |>
  group_by(treatment) |>
summarize(
  mean_therm1 = mean(therm1, na.rm = TRUE),
  sd_therm1 = sd(therm1, na.rm = TRUE))
```

```
# A tibble: 2 x 3
  treatment                                 mean_therm1 sd_therm1
  <chr>                                           <dbl>     <dbl>
1 No Contact                                       57.9      28.7
2 Same-Sex Marriage Script by Gay Canvasser        59.4      28.4
```

```r
q33
```

```
# A tibble: 2,441 x 6
   study treatment                                therm1 therm2 therm3 therm4
   <dbl> <chr>                                     <dbl>  <dbl>  <dbl>  <dbl>
 1     2 No Contact                                    2      6      0      6
 2     2 No Contact                                   91     90    100     93
 3     2 Same-Sex Marriage Script by Gay Canvasser     52     58     53     59
 4     2 Same-Sex Marriage Script by Gay Canvasser     43     NA     31     48
 5     2 Same-Sex Marriage Script by Gay Canvasser    100     94    100     NA
 6     2 Same-Sex Marriage Script by Gay Canvasser     92     97     87     93
 7     2 Same-Sex Marriage Script by Gay Canvasser     50     46     NA     47
 8     2 Same-Sex Marriage Script by Gay Canvasser     99    100     92    100
 9     2 No Contact                                   46     67     38     46
10     2 Same-Sex Marriage Script by Gay Canvasser     50     45     53     51
# i 2,431 more rows
```

```r
(59.35057   - 57.88612) / sqrt((28.41578^2 + 28.69655^2)/2)
```

```
[1] 0.05128253
```

- Mean No Contact = 57.88612
- Standard Deviation No Contact = 28.69655
- Mean Gay Canvasser = 59.35057
- Standard Deviation Gay Canvasser = 28.41578

The standardized difference in means is about 0.05. Because it is below 0.25, it suggests a good balance.

```r
## Q3.4

q33 |>
  group_by(treatment) |>
summarize(
  mean_therm2 = mean(therm2, na.rm = TRUE),
  sd_therm2 = sd(therm2, na.rm = TRUE),
  n = sum(!is.na(therm2)))
```

```
# A tibble: 2 x 4
  treatment                                mean_therm2 sd_therm2     n
  <chr>                                          <dbl>     <dbl> <int>
1 No Contact                                      57.6      28.6  1039
2 Same-Sex Marriage Script by Gay Canvasser       61.2      28.8  1093
```

```r
61.20311 - 57.63523
```

```
[1] 3.56788
```

The mean post-period thermometer score is higher in the gay canvasser group, with a difference of roughly 3.57 points compared to the no-contact group.

```r
##Q3.5

SE <- sqrt((28.75285^2/1093) + (28.58670^2/1039))

SE
```

```
[1] 1.242138
```

```r
diff <- 3.56788

lower <- diff - 1.96*SE

upper <- diff + 1.96*SE

lower
```

```
[1] 1.133289
```

```r
upper
```

```
[1] 6.002471
```

The 95% CI is roughly [1.13, 6.00]. Since zero is **not** in the interval, we reject no difference and conclude that the gay canvasser group has higher average *therm2* than the no-contact group after the conversation. In other words, the groups differed after the conversations.

```
#3.6

study2_control <- gay_r |>
  filter(study ==2, treatment == "No Contact")

tibble(
  wave = c("therm2", "therm3", "therm4"),
  correlation_with_term1 = c(
cor(study2_control$therm1, study2_control$therm2, use = "complete.obs"),
cor(study2_control$therm1, study2_control$therm3, use = "complete.obs"),
cor(study2_control$therm1, study2_control$therm4, use = "complete.obs")))
```

```
# A tibble: 3 x 2
  wave   correlation_with_term1
  <chr>                   <dbl>
1 therm2                  0.973
2 therm3                  0.959
3 therm4                  0.971
```

The correlations remain very high but decline slightly as the waves move further from the baseline. Therm 1 and therm 2 have the highest correlation, then therm1 and therm4, and lastly therm 1 and therm 3. Thus, this suggests that the real survey responses are fairly stable over time (although there is some variation).

### Q3.7

Study 2's data is definitely more realistic compared to Study 1's. In Study 1, the re-test correlation between therm1 and therm2 was extremely high (roughly 0.998 and the standard deviation of within-person change was only about 1.99, meaning that responses barely changed over time. Study 2, on the other hand, had a lower and more realistic correlation (about 0.97) and much larger standard deviation (6.6), meaning that people's responses varied more between survey waves (which is more realistic). The treatment effect in Study 2 was also modest and messier, with the gay canvasser group averaging about 3.6 thermometer points higher than the no-contact group, which is more aligned with real data. It makes sense that real treatment effects would be messy because people's opinions fluctuate and survey responses are affected by measurement error, such as interpretation or uncertainty. Simple tools like means, standard deviations, histograms, and correlations helped detected the fraud because they showed patterns in Study 1 that were unrealistically consistent compared to real survey data.